



The impact of business process orientation on financial and non-financial performance

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Abstract

Purpose – Extensive literature on business process management suggests that organizations could enhance their overall performance by adopting a process view of business. However, there is a lack of empirical research in this field. The purpose of this paper is to investigate the understanding of the process view and process maturity levels in a transition economy and to test the impact of process orientation maturity level on organizational performance.

Design/methodology/approach – Empirical investigation combined an exploratory-confirmatory approach using factor analysis and structural equation modeling.

Findings – The investigation confirms the impact of business process orientation on organizational performance in a transition economy. The link is even stronger than in the original investigation. The results show that business process orientation leads to better non-financial performance and indirectly to better financial performance.

Practical implications – The research confirms that business process orientation is advantageous for companies since it has a positive influence on organizational performance. The finding that the impact on financial performance is indirect through non-financial performance suggests that the companies have to take that view of performance into consideration as well.

Originality/value – The paper is valuable for academics and practitioners because the impact of business process orientation on organizational performance has been confirmed for a transitional economy. Its originality is in the measurement of organizational performance, for which a more detailed specification of organizational performance based on the balanced scorecard concept that includes non-financial performance measures has been used.

Keywords Process management, Organizational performance, Mathematical modeling

Paper type Research paper



Introduction

Organizations are continually under competitive pressures and forced to re-evaluate their business models and underlying business processes. Business processes represent a core of the functioning of an organization because the company primarily consists of processes, not products or services. In other words, managing a business means managing its processes (McCormack and Johnson, 2001). Despite their importance, business processes have been neglected in managerial studies for a long time mainly due to the fact that departments in companies are structured in a functional or product oriented way (Vanhaverbeke and Torremans, 1998).

Extensive literature on business process management (Davenport, 1993; Hammer and Champy, 1993; McCormack and Johnson, 2001; Burlton, 2001; Harmon, 2003) suggests that organizations can enhance their overall performance by adopting a process view of business. To analyze the level of adoption the concept of business process orientation can be used. Business process orientation is extremely important for the success of business process management efforts within organizations and its lack can result in all kinds of problems in that regard (Reijers, 2006). However, there has not been much empirical research on the impact of business process management practices and business process orientation on organizational performance. Nevertheless, some of the previous studies (McCormack and Johnson, 2001) confirmed that the companies, which have reached a higher level of business process maturity consistently outperform those that have not reached it.

Based on the original study (McCormack and Johnson, 2001) a joint empirical research by the Faculty of Economics of Ljubljana and Zagreb was carried out in Slovenia and Croatia. The aim was to investigate the understanding of the process view and process maturity levels of Slovenian and Croatian companies (Škrinjar *et al.*, 2006) and to test the impact of a process orientation maturity level on organizational performance. The paper presents the results of the empirical research that confirms the impact of business process orientation on organizational performance. The contribution of our study is the verification of the link in a transition economy and a more detailed specification of organizational performance that includes non-financial performance measures.

The paper is structured as follows: the next section reviews the relevant literature on business process orientation, organizational performance and the link between the two concepts. In the third section, the research model is conceptualized and research hypotheses are presented. Section 4 aims at presenting a methodological framework for the study and provides the results of data analysis. Section 5 concludes with a summary of the main findings, discusses them from theoretical and practical standpoints, and outlines directions for future research together with the limitations of the study.

Literature review

Business process orientation

Although definitions of the business process orientation (BPO) vary, we adopt the McCormack and Johnson (2001) definition of process orientation: An organization that, in all its thinking, emphasizes process as opposed to hierarchies with a special emphasis on outcomes and customer satisfaction. Process orientation still is not recognized as an independent discipline; rather it represents a generic concept of numerous management philosophies which use process perspective to improve business performance (Lindfors, 2003).

It becomes evident that all organizations are composed of business processes, even though they have not adopted the process view. Therefore, a process approach can be applied to each organization and the level of process orientation can be determined with respect to business performance. The levels of process orientation are often presented by a process maturity concept. In the current business environment, there is no scarcity of process maturity models (Škrinjar *et al.*, 2006). They serve as reference models of the stages that organizations go through as they move from being immature

to mature in their process orientation. The business process maturity models are based on concepts developed by researchers over the past two decades and imply that a process has a lifecycle that is assessed by the extent to which the processes are explicitly defined, managed, measured and controlled.

For the purpose of this research, the BPO maturity model was readjusted from McCormack and Johnson (2001). The original model was developed based on the concepts of process maturity, BPO, and the Capability Maturity Model developed by the Software Engineering Institute at Carnegie Mellon University (Lockamy III and McCormack, 2004). The BPO construct describes a four-step pathway for systematically advancing business processes along the maturity continuum (*Ad Hoc*, *Defined*, *Linked*, and *Integrated* level). Each step builds on the work of the previous steps to apply improvement strategies that are appropriate for the current maturity level. The following definitions are provided for the stages that an organization goes through when becoming business process oriented:

- (1) *Ad Hoc*: The processes are unstructured and ill-defined. Process measures are not in place and the jobs and organizational structures are based upon traditional functions, not horizontal processes.
- (2) *Defined*: The basic processes are defined and documented and are available in flow diagrams. Changes to these processes must now undergo a formal procedure. Jobs and organizational structures include a process aspect, and yet remain basically functional. Representatives from functional areas (sales, manufacturing, etc.) have regular meetings to coordinate with each other, but only as representatives of their traditional functions.
- (3) *Linked*: The breakthrough level. Managers employ process management with strategic intent and results. Broad process jobs and structures are put in place outside the traditional functions.
- (4) *Integrated*: The company, its vendors and suppliers, take cooperation to the process level. Organizational structures and jobs are based on processes, and traditional functions begin to be equal or sometimes subordinate to the process ones. Process measures and management systems are deeply imbedded in the organization (McCormack and Johnson, 2001; McCormack, 2003).

The BPO construct is measured at an aggregate level and by the following individual dimensions: process view, process jobs, process management, and measurement. Process view involves a focus on the workflows and processes across an organization. Functional roles and titles reflecting the traditional hierarchical structure are replaced by process owners-leaders who are responsible and accountable for the operation and improvement of the core business (Tenner and DeToro, 1996; Ostroff, 1999; Harmon, 2003). Along with process owners, process teams become the main building element of an organization, which emphasizes the process job category. Finally, process management, supplemented with process measures, has as much to do with changing the culture and the way of thinking, as it does with setting process oriented measures and goals for an organization.

Organizational performance

Organizational performance comprises the actual output or results of an organization as measured against its intended outputs: goals and objectives. Organizational performance measures allow companies to focus attention on areas that need

improvement by assessing how well work is done in terms of cost, quality, and time. With the pressure of world class competition, organizational performance measurement has become increasingly necessary for the continued survival of today's companies. Therefore, there is an extensive amount of literature on performance measurement, frameworks and systems. Issues concerning organizational performance can be divided into two main fields:

- (1) performance measures; and
- (2) performance frameworks and systems.

Attempts have been made in the past to measure performance based on quantitative financial measures, while less emphasis has been placed on the qualitative components of performance measurement. Hence, Maskell (1992) suggests that performance measures should primarily use non-financial performance techniques and change overtime as the company needs change. It is also important to involve qualitative indicators, such as customer service and satisfaction, product quality, learning and innovation (Kaplan and Norton, 1996; Neely, 2002; Neely *et al.*, 2002). According to Waggoner *et al.* (1999) performance measures within an organization can be designed on the basis of six different approaches:

- (1) the engineering approach which measures the input/output ratio;
- (2) the system approach which sets objectives for each work unit and measures the achievement of these objectives;
- (3) the management accounting approach measuring the achievement of financial results;
- (4) the statistical approach which extends the engineering approach by providing empirically tested information about input/output processes;
- (5) the consumer marketing approach which measures consumer satisfaction; and
- (6) the "conformance to specifications" approach which advocates the use of a checklist of attributes of a product or service and its service delivery system.

In order to achieve business excellence, it is necessary for an organization to develop a system for performance measurement. To address this issue, an interdisciplinary review of organizational performance measurement frameworks is espoused in both academic literature and business press (Waggoner *et al.*, 1999; Kuwaiti and Kay, 2000; Kutucuoglu *et al.*, 2002; Folan and Browne, 2005; Lin and Chen, 2007). One cannot evaluate organizational performance without taking organizational goals into consideration. The modern business environment demands a multi-goal orientation. Profit theory (Cyert and March, 1963) is no longer a valid measure of organizational performance and neither are other approaches that only take the interests of shareholders (owners) of a company into account. Indeed, Agrell *et al.* state that "although a profit maximizing behavior may be induced by profit sharing schemes, such contracts may easily lead to sub-optimal levels of organizational training, innovation, and knowledge transfer" (Agrell *et al.*, 2002). Today's business environment is characterized by the increasing importance and strength of various stakeholder groups.

It has become quite obvious that all stakeholders need to be taken into account when assessing modern company's performance. This is the main idea of Freeman's

Stakeholder theory (Freeman, 1984, 1994). The stakeholder view maintains that firms have stakeholders rather than just shareholders to account for. The view that the corporation has obligations only to its stockholders is replaced by the notion that there are other groups to whom the firm is also responsible. Groups with a “stake” in the firm include shareholders, employees, customers, suppliers, lenders, the government, and society (Berman *et al.*, 1999; Harrison and Freeman, 1999; Hillman and Keim, 2001; Riahi-Belkaoui, 2003).

Behavioral theory (Cyert and March, 1963) recognized the company as a coalition of individuals or groups of individuals such as management, employees, customers, owners, government, etc. but did nothing to introduce this finding into organizational performance assessment. Emerging management paradigms emphasize a stakeholder perspective (Atkinson *et al.*, 1997; Berman *et al.*, 1999; Harrison and Freeman, 1999; Hillman and Keim, 2001; Sirgy, 2002; Riahi-Belkaoui, 2003; Tangem, 2004).

One important notion revealed in many studies is that building better relations with primary stakeholders like employees, customers and suppliers could lead to increased shareholder’s wealth. A sustainable organizational advantage may be built with tacit assets that derive from developing relationships with key stakeholders (Hillman and Keim, 2001). When studying the relationship between stakeholder management and a firm’s financial performance, Berman *et al.* (1999) found that fostering positive connections with key stakeholders (customers and employees) can help a firm’s profitability.

Due to the significance of various stakeholders, organizational performance should not be solely assessed by financial indicators. There are several approaches (Tangem, 2004) to organizational performance measurement that encompass different stakeholder’s perspectives. The balanced scorecard (BSC) (Kaplan and Norton, 1992, 1993, 1996) is the most established and most commonly used (Neely, 2005), but by far not the only one. The multi-model performance framework (MMPF) model by Weerakoon (1996) is also very interesting and has four-dimensions including employee motivation, market performance, productivity performance, and societal impact, and covers the satisfaction of various stakeholders such as customers, investors, employees, suppliers, and society. A more recently developed conceptual framework is the performance prism, which suggests that a performance measurement system should be organized around five distinct but linked perspectives of performance (Tangem, 2004, p. 733).

Link between BPO and organizational performance

This paper explores the association between BPO and organizational performance. The goal of the research is to extend the limited literature on BPO projects and organizational performance measures and frameworks. Similarly, a research was conducted by Sharma (2005) to provide the evidence that ISO 9000 certification is associated with improvements in financial performance. Specifically, the multivariate test showed that ISO 9000 certification is associated with significant improvements in profit margin, growth in sales, and earning per share. The results showed that the effect was greater on profit margin than on the growth in sales. This suggested that the improvement in overall performance was largely attributed to improvements in internal business processes. Martinez Sanchez *et al.* (2007) conducted a research to explore the relationship between teleworking adoption, workplace flexibility and

firm performance. The results indicated that firm performance was positively related, which suggested that organizational change increased organizational flexibility and generated sustainable competitive advantage.

Theoretically, the benefits that BPO brings to an organization are numerous. It affects the soft side of organizations as well as the bottom line figures. Some of the benefits reported in the literature are: cost savings through a more efficient execution of work, reduced cycle times, improved customer focus, better integration across the organization, increased flexibility of the firm along with improved customer satisfaction, elimination of redundant, and duplicated activities (Keen, 1997; Sikavica and Novak, 1999; Oden, 1999, Galbraith, 2002). Implementing BPO as a way of organizing and operating in an organization will improve internal coordination and break down the functional silos that exist in most companies (McCormack and Johnson, 2001).

While the literature presents more than a plentiful supply of benefits of BPO, their empirical confirmation is scarce. In their seminal work McCormack and Johnson (2001) conducted an empirical study to explore the relationship between BPO and enhanced business performance. The research results showed that BPO is critical in reducing conflict and encouraging greater connectedness within an organization, while improving business performance. Their results indicate a surprisingly strong relationship between BPO and overall performance. Furthermore, the more business process oriented an organization is, the better it performs both from an overall perspective as well as from the perspective of the employees. Considering all the factors that can potentially affect business performance, this finding is compelling (McCormack and Johnson, 2001).

Research hypotheses and model conceptualization

The main purpose of this paper is to test if higher levels of business process orientation lead to better organizational performance. While a similar research was carried out by McCormack and Johnson (2001) our contribution is the application of the link in a transition economy and a detailed specification of organizational performance. Since the theoretical background of the constructs has been presented in the first part of the paper, this section presents the hypothesized relationships and the rationale for them.

As companies renovate themselves and change existing practices and adopt new ones striving to become more process oriented they inevitably optimize their processes and organizational structures that support them. There is an abundance of literature (Davenport, 1993; Hammer and Champy, 1993; McCormack and Johnson, 2001; Burlton, 2001; Harmon, 2003) arguing that renovation of business processes will be translated into better financial performance. We adopt this view and based on it we present our first hypothesis:

- H1.* The higher level of business process orientation a firm achieves the better it performs financially.

Becoming more process oriented has a profound impact on many facets of an organization. It changes the way employees work and interact. As functional silos are broken down and business processes start to get integrated inter-functional conflicts decrease and inter-departmental connectedness increases. Furthermore, becoming more process oriented strengthens esprit de corps (McCormack and Johnson, 2001) all of which has a positive effect on the satisfaction of employees (Reijers, 2006). Process orientation also changes the interaction between a firm and its business partners (suppliers and customers)—by integrating processes beyond the boundaries of a firm

transaction based cooperation is transformed into long-term partnership that results in increased performance for all links in a supply chain (Cousins and Menguc, 2006; Hendricks *et al.*, 2007). In this context, we stipulate our second hypothesis as follows:

H2. The higher the level of business process orientation a firm achieves, the better it performs non-financially in terms of more satisfied employees, customers and suppliers.

The rationale for our third hypothesis stated below is as intuitive as is widely supported in literature. Firstly, satisfied employees perform better and execute their tasks more effectively and efficiently. They are less inclined to change jobs (Karatepe *et al.*, 2006) and are less frequently absent all of which should have a positive impact on financial performance. Satisfied customers make repeat purchases and purchase more. Good, long-term relationship with suppliers is beneficial for both parties involved (Hendricks and Singhal, 2003; Dehning *et al.*, 2004). Therefore:

H3. Better non-financial performance leads to better financial performance.

In Figure 1 the conceptualized model along with the hypothesized relationships is shown.

Research methods and data analysis

Research instrument

In order to test the proposed hypotheses, our instrument was composed of two parts. The first part measuring BPO was adopted from the original study conducted by McCormack and Johnson (2001). This part of the instrument was amended in two sections:

- (1) one question was added to the Process view section; and
- (2) two questions were added to the Process management and measurement section.

Altogether BPO construct was measured by using 15 items. Even though the original instrument from McCormack and Johnson included an overall organizational performance construct, it was only measured by two items. As our goal was to tap deeper into the problem we chose not to use this operationalization of organizational performance construct. Therefore, we devised our own instrument for organizational performance measurement. This represents the second part of our research instrument. Theoretically, it is based on the BSC (Kaplan and Norton, 1992, 1993, 1996) and its main goal is to measure different facets of

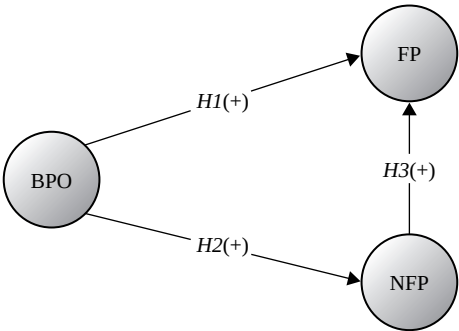


Figure 1.
Conceptual model and
direction of impact
between constructs

organizational performance, namely the financial and the non-financial. We used 19 items to measure the organizational performance. The basic structure and nature of the research instrument can be seen in Tables I and II.

In addition to the latent variables measurement items, some general questions for basic descriptive statistics were also included in the questionnaire.

To test the hypothesized relationships, we employed the combined exploratory-confirmatory approach. In the first phase, we analyzed the questionnaire items by using the exploratory factor analysis in order to test if the items in fact measure pre-specified constructs. In the second phase, following the approach proposed by Diamantopoulos and Siguaw (2000), we tested the relationships between constructs by using the structural equation modeling.

Data collection and sample characteristics

During September and October 2005 questionnaires were distributed to Slovenian and Croatian companies with more than 50 employees. The main source of data about Slovenian companies was The Business Directory of Slovenia (IPIS). The questionnaire was sent to all 1237 companies of the defined size across industries. In Croatia, data were gathered from the database of The Institute for Business Intelligence and the questionnaire was sent randomly to the half of the population, which counts for 1,750 companies. In both countries, the questionnaire was addressed to the CEOs or the chairpersons of the companies who were instructed to fill in the questionnaire themselves or give it to a competent person within the organization.

In the Slovenian case, 203 completed questionnaires were returned (which accounts for 16.5 per cent response rate) while in Croatia 202 completed questionnaires were

	1	Factor 2	3
procv1–The average employee views the business as a series of linked processes		0.504	
procv2–Process terms such as input, output, process, and process owners are used in conversation in the organization		0.496	
procv3–Processes within the organization are defined and documented using inputs and outputs to and from our customers		0.709	
procv4–The business processes are sufficiently defined so that most people in the organization know how they work		0.691	
procv5–Implementation of information technology is based on the processes, not on functions		0.600	
procj1–Jobs are usually multidimensional and not just simple tasks			0.729
procj2–Jobs include frequent problem solving			0.436
procj3–People are constantly learning new things on the job			0.447
procmm1–Process performance is measured in the organization	0.806		
procmm2–Process measurements are defined	0.860		
procmm3–Resources are allocated based on process	0.551		
procmm4–Specific process performance goals are in place	0.686		
procmm5–Process outcomes are measured	0.665		
procmm6–An on-line control of information quality in processes has been established	0.472		
procmm7–Information flow through process is continuous and efficient	0.402		

Table I.
Rotated factor Matrix
for business process
orientation

Table II.
Rotated factor Matrix for
organizational
performance

	Factor			
	1	2	3	4
FINA1–ROA				0.738
FINA2–value added per employee				0.773
NFSUP–relationships with suppliers		0.457		
NFEMP1–net fluctuation of employees	0.433			
NFEMP2–work productivity of employees			0.588	
NFEMP3–trust into leadership	0.713			
NFEMP4–employees mutual trust	0.669			
NFEMP5–work organization of employees	0.659			
NFEMP6–commitment of employees	0.746			
NFEMP7–employees prepared to go extra mile	0.584			
NFEMP8–costs of work			0.506	
NFEMP9–absenteeism			0.469	
NFEMP10–satisfaction with work conditions	0.689			
NFEMP11–learning ability and adaptability	0.607			
NFEMP12–risk taking				
NFCUST1–customer complaints number		0.674		
NFCUST2–customer complaints speed of solving		0.644		
NFCUST3–loosing/retaining clients		0.734		
NFCUST4–reputation of a company in customers eyes		0.689		

returned to the research group (which accounts for 11.5 per cent response rate). The descriptive statistics of our sample are presented in Table III.

Exploratory factor analysis

Before testing the entire model by using the structural equation modeling, we conducted an exploratory factor analysis to get a first insight into our data and to assess the validity of our measurement model. The main concern in this part is “Do items really measure the specified constructs?” We used the statistical package SPSS 13.0 to run a series of data reduction tests. Data were subjected to Factor Analysis Technique using the principal axis factoring extraction method combined with Varimax rotation.

Firstly, we analyzed the items measuring the business process orientation construct. As this construct had been tested extensively, the results shown in Table I were anticipated as three factors emerged, each representing one aspect of BPO. Using 0.50 loading cut-off value, which according to Comrey and Lee (1992) is a good score, two things need to be pointed out:

- (1) out of three additional items that were amended to the initial instrument, one (procv5) has loaded appropriately on its factor; factor loading for item procm6 was very close to cut-off value so we decided to keep it in the analysis; factor loading for item mmproc7 was well below the cut-off value and was therefore omitted from further analysis; and
- (2) though two items loadings on the second factor have not reached the 0.50 value, we have decided to keep them in our analysis as their loading was very close to the prescribed one and have already been validated in the original study.

	Croatia (per cent)	Slovenia (per cent)	Business process orientation
<i>Independent characteristics of the companies</i>			
<i>Number of employees</i>			
50-249	71.3	72.4	
250-499	16.3	16.1	
500 +	11.8	11.5	
Not given	0.5	–	
<i>Annual revenue in 2004 (mil. €)</i>			
Under 7	50.0	37.4	
7-27	33.7	42.8	
More than 27	14.4	19.8	
Not given	2.0	7.9	
<i>Independent characteristics of the respondents</i>			
<i>Hierarchical position of the respondents</i>			
Top management	22.4	31.5	
Middle management	55.7	31.5	
Lower management and operational level	13.0	12.8	
Not given	8.9	24.2	

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Table III.

Sample characteristics

Our analysis continued with organizational performance measurement items. Closely reflecting our expectations, the factor analysis of OP measurement items revealed four factors (Table II). The fourth factor clearly deals with financial performance. Factors one (subjective measures) and three (objective measures) tap into non-financial performance. More precisely into employee perspective of non-financial performance and the second factor represents the customer perspective of non-financial performance. Considering the 0.5 cut-off loading value, the following items were removed from further analysis: NFEMP1, NFEMP9, and NFEMP12. Even though the item NFSUP did not load on its own factor, which is probably due to the fact that we only used one item for measuring the relationship with suppliers, and had the loading value below 0.5 we decided to keep the item in further analysis as this aspect of non-financial performance is firmly grounded in theory (Freeman, 1984, 1994).

Operationalization of business process orientation and organizational performance

After subjecting the data to factor analysis, the purified data can now be used to operationalize the measurement of BPO and OP constructs. As the BPO construct has already been operationalized by McCormack and Johnson, we have closely followed their route and grouped items into three scales:

- (1) process view (PROCV) includes items procv1 through procv5;
- (2) process jobs (PROCJ) includes items procj1 through procj3; and
- (3) process management and measurement (PROCMM) includes items procmm1 through procmm6.

Based on extensive literature review and our intention to scrutinize organizational performance in detail, organizational performance is included in the model as two separate constructs: financial performance (FP); and non-financial performance (NFP). The construct FP is measured by using FINA1 (measuring ROA) and FINA2

(value added per employee). Construct NFP is measured using items grouped into four scales: (1) NFSUP—one item (nfsup); (2) NFEMP1—seven items (nfemp3 through nfemp7 and nfemp10 through nfemp11); and NFEMP2—two items (nfemp2 and nfemp8); NFCUST—four items (nfcust1 through nfcust4).

Confirmatory analysis using structural equation modeling

The next step in our analysis was the assessment of model fit. Here, we were interested in whether the hypothesized model is consistent with the data. First, we examined the measurement part of the model. Our aim was to determine the validity and reliability of the measures used to represent the constructs of interest. Validity reflects the extent to which an indicator actually measures what it is supposed to measure. The validity can be assessed by examining the magnitude and significance of the loading paths λ that represents a direct relationship between the indicator and the construct. All λ 's should be significant (t -values should exceed 1.96) and exceed 0.50 threshold (Hair *et al.*, 1998; Prajogo and McDermott, 2005). As it can be seen from the Table IV, all indicator loading values are significant (at $p < 0.01$ or better— t -values exceed 2.64) and exceed 0.50, which provides validity evidence in favor of the indicators used to represent the constructs of interest.

To test the composite (construct) reliability, the composite reliability index (CRI) and average variance extracted (AVE) were calculated. Composite reliability assumes that a set of latent construct indicators is consistent in the measurement (Škerlavaj *et al.*, 2006). There is no generally acceptable standard for adequate values of CRI. Koufteros (1999) suggested values above 0.80, while Diamantopoulos and Siguaw (2000) were satisfied with 0.60. AVE is similar to CRI with the one exception that standardized loadings are squared before they are summed (Hair *et al.*, 1998; Koufteros, 1999). The cut-off value most often used for AVE is 0.50 (Bagozzi and Yi, 1988; Hair *et al.*, 1998), while there are also cases where a milder restriction of 0.40 was employed (Diamantopoulos and Siguaw, 2000). As it can be seen from Table V, all values for CRI and AVE surpass the prescribed values; therefore the reliability of indicators is acceptable.

Before examining the hypothesized relationships, the global fit of the model needs to be assessed. There is a plethora of goodness-of-fit indices that can be used as summary measures of a model's overall fit (Diamantopoulos and Siguaw, 2000). Research evidence

Table IV.
Unstandardized,
completely standardized
loading estimates and
 t -values

		LAMBDA-X		t -value
		Unstandardized factor loading	Completely standardized factor loading	
BPO	PROCV	1.00	0.85	— ^a
	PROCJ	0.55	0.52	8.63
	PROCMM	0.87	0.68	10.17
		LAMBDA-Y		
NFP	FINA1	1.00	0.70	— ^a
	FINA2	1.24	0.95	9.84
	NFSUP	1.06	0.66	13.41
FP	NEFEM1	1.00	0.85	— ^a
	NEFEM2	0.79	0.66	13.35
	NEFCUST	0.92	0.72	14.79

Note: ^a Indicates a parameter fixed at 1.00 in the original solution

supports the need to use more than one index (Breckler, 1990; Bollen and Long, 1993; Tanaka, 1993; Coenders *et al.*, 2003). Comparative fit index (*CFI*—should exceed value of 0.90) and non-normed fit index (*NNFI*; also named the Tucker-Lewis fit index: *TLI*—should exceed value of 0.90) are used most often to assess model fit (Koufteros, 1999). In addition to these, we also used root mean square error of approximation (*RMSEA*—values between 0.05 and 0.08 are indicative of a reasonable), standardized root mean square residual (standardized *RMR*—values below 0.05 are indicative of a good fit) and goodness of fit index (*GFI*—should exceed value of 0.90) for additional assessment of model fit.

The results of fitting the model to the data show that the model had a good fit as values for all of the most frequently used indices fall within the acceptable range: *RMSEA* = 0.058; *NNFI* = 0.96; *CFI* = 0.97; standardized *RMR* = 0.039; *GFI* = 0.97. In Figure 2, the path diagram of our model is presented.

As the overall model exhibited a good fit, the structural part can be examined. The aim is to determine whether the theoretical relationships specified by our hypotheses are indeed supported by the data. Three issues are of relevance here. First, we examine if the signs of the parameters representing the paths between the constructs indicate the same direction as hypothesized. In that regard only Hypotheses 2 and 3 are supported by the data as there is a positive sign between BPO and NFP and between NFP and FP. Hypotheses 1 is not supported as the direction of the relationship

	CRI	AVE
BPO	0.73	0.48
FP	0.81	0.69
NFP	0.82	0.53

Table V.
Composite reliability
index and average
variance extracted

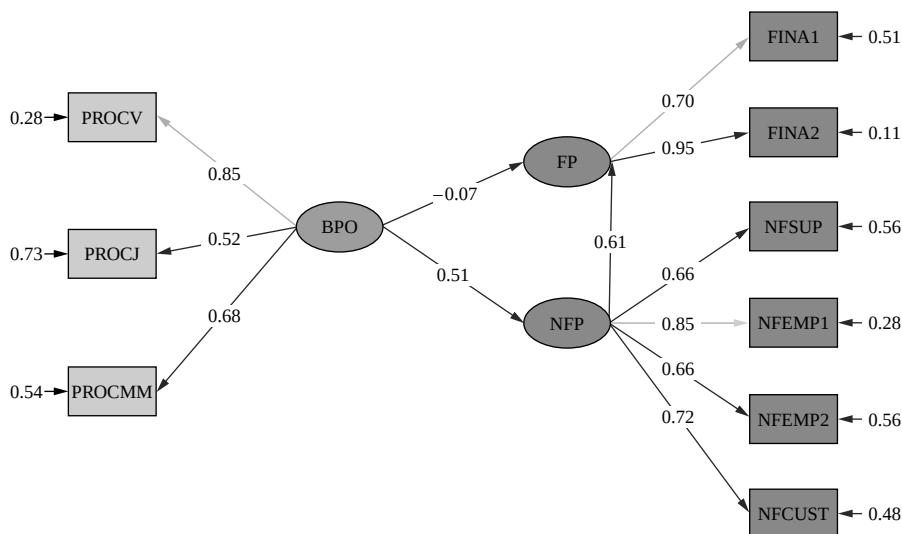


Figure 2.
Path diagram of
conceptualized model

in not as stated. Second, we examine the statistical significance of parameters. Again only paths representing second and third hypotheses were found to be statistically significant (t values: 7.70 and 7.05, respectively). The relationship between BPO and FP was not found to be statistically significant. Third, the squared multiple correlations (R^2) for the structural equations are inspected as they indicate the amount of variance of endogenous constructs that is accounted for by independent constructs. For the hypotheses 2 and 3 the R^2 s are high (0.26 and 0.33, respectively) indicating a strong relationship. Again the lack of direct relationship as stated in Hypothesis 1 was reconfirmed in very low R^2 (0.057) for the path representing that link.

Considering all three aspects of the structural relationship we accept the second and the third hypothesis but reject the first one.

Discussion and implications

The main goal of our study was to determine whether higher levels of business process orientation lead to better organizational performance in companies in transitional economies. The data from the empirical study, which have been subjected to rigorous statistical techniques, have shown support for that. Therefore, based on our hypotheses we conclude that higher levels of business process orientation lead to better financial and non-financial performance in these circumstances. Furthermore, it has been shown that there is a strong direct impact of BPO on non-financial performance. On the other hand, no such impact has been found between BPO and financial performance. It does not mean there is no connection whatsoever. It has been shown that BPO has a strong indirect impact on financial performance through non-financial performance.

The contribution of our study is two-fold: first the extension of the original study (McCormack and Johnson, 2001) whereby we have scrutinized the effects of BPO on OP in much more detail by capturing and analyzing performance in coherence with the stakeholder theory (Freeman, 1984) and BSC approach (Kaplan and Norton, 1992, 1993, 1996). We did that by inclusion of key stakeholders (customers, employees and suppliers) in assessment of non-financial performance in addition to financial performance. Our conceptualization of the model therefore included two distinct yet related constructs of organizational performance. The extension of the organizational performance was made by significantly expanded measurement instrument for capturing organizational performance. This approach has proved to be useful as we were able to decompose the impact of BPO on OP and reveal a deeper structure of the relationship. Second, we carried out the study in a transition economy and found that the original findings are also applicable in this socio-economic environment.

The results of our study also have many practical implications for managers. As companies renovate themselves and adopt new practices striving to attain higher levels of process orientation, it enables them to improve their relationship with its key stakeholders employees, customers, and suppliers by creating a fertile environment for conducting business. Clearly such environment is also a catalyst for better financial performance. Therefore, as business environment gets more competitive, business process orientation offers a way to adapt to new conditions and circumstances. Since higher business process orientation maturity levels lead to better organizational performance, managers need to familiarize themselves with this concept and practical implementation issues. They need to examine their current practices, structures and

management and measurement processes and assess the current state. BPO maturity model then serves as a road map for their improvement and renovation efforts.

Though we rejected the first hypothesis postulating the positive effect of BPO on financial performance, this of course is not conclusive. The lack of a significant effect might be due to time lag in the effects. Clearly, organizational renovation and process improvement practices cannot and do not happen instantaneously but take a lot of time. Financial results of such efforts might not arise immediately. This points to one limitation of our research, namely the study has not been devised in a longitudinal nature. Therefore, time lag in effects cannot be detected. This dimension is left for future research. For analyzing the time lag in effects and for gaining wider generalization ability, the study will be repeated in Slovenia and Croatia and extended to new countries.

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Further reading

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